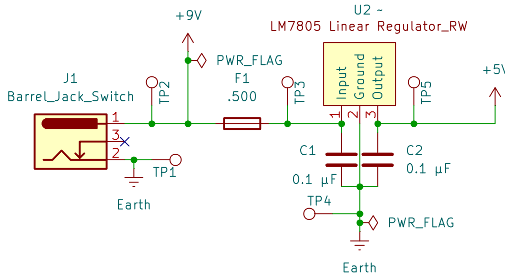
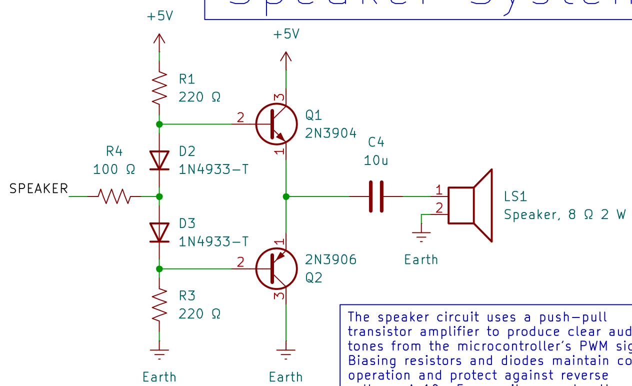
Power Supply



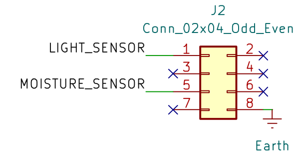
The board is powered through a 5 V DC barrel jack. The input passes through a 200–250 mA fuse for overcurrent protection and includes decoupling capacitors to filter noise and stabilize the supply. The LM7805 voltage regulator footprint is available but left unpopulated when a regulated 5 V adapter is used. A 10 µF electrolytic capacitor provides bulk decoupling, and 0.1 µF ceramic capacitors are placed near integrated circuit power pins. Optional protection components such as a Schottky diode for reverse polarity and a transient voltage suppressor can be added near the barrel jack if needed. Test points for the 5 V rail and ground are included for easy measurement and debugging.

Speaker System



The speaker circuit uses a push-pull transistor amplifier to produce clear audio tones from the microcontroller's PWM signal. Biasing resistors and diodes maintain correct operation and protect against reverse voltage. A 10 µF capacitor couples the signal to the speaker, blocking DC while passing the audio waveform. The design is optimized for small 8 Ω speakers, providing an efficient and simple alert or tone output suitable for low-power applications.

Ribbon Connector

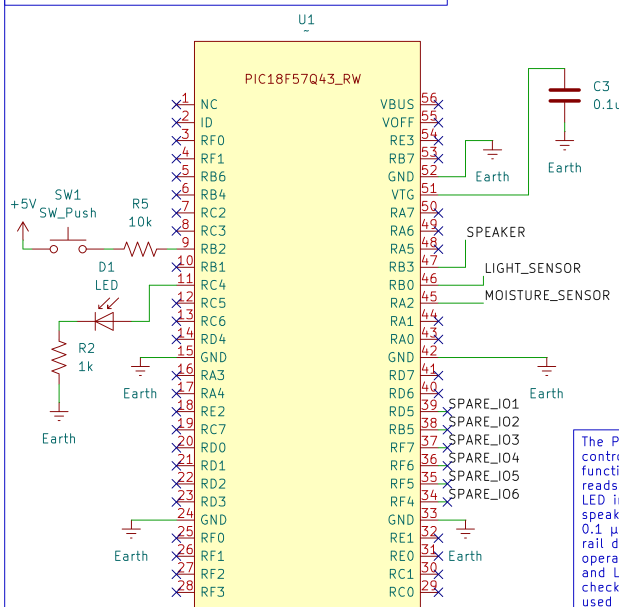


An 8-pin ribbon connector provides communication between this board and the hub. Pin 1 is unused, and pin 2 carries the digital alert signal to the hub. Pins 3 through 5 are reserved for future use. Pin 6 brings an analog signal into the board, routed through the op-amp buffer and read by RA0. Pin 7 carries an analog or DAC output signal from RA2 to the hub, and pin 8 is the shared system ground. Each digital line that leaves the board includes a 100 Ω series resistor to protect against transients or accidental contention. No 5 V power is shared across the ribbon.

Expansion and Design Intent

Unused microcontroller ports such as RA1 and other RC, RD, or RF pins are labeled for future I/O expansion. Extra channels of the MCP6004 are available for additional analog signal conditioning. The layout and connector pinout follow the Irri-Gators team interface standard to ensure compatibility between subsystems. All unused pins are clearly marked with either "Future I/O" or "No Connect" symbols. The board's design allows for simple integration of new components or communication features without schematic restructuring.

Microcontroller



The PIC16F870A3 Curiosity Nano controls local input and output functions for the subsystem. It reads sensor data, manages the LED indicator, and drives the speaker through a PWM output. A 0.1 µF capacitor provides power rail decoupling for stable operation, while the pushbutton and LED allow quick functional checks. Analog-capable pins are used for the light and moisture sensors, and short trace routing minimizes noise. Proper grounding and decoupling ensure reliable microcontroller performance during operation.

General Notes

All resistors are quarter watt unless otherwise specified, and all capacitors are listed in microfarads unless otherwise noted. Power traces are designed with at least 40 mil width to safely handle the system current. Component values such as op-amp gain resistors and buzzer coupling capacitors may be adjusted during testing to achieve the desired audio performance. The board is electrically isolated from the hub except for signal and ground connections. This schematic represents version 1.0 of the Audio and Alert subsystem, completed in October 2025.

Testing

The board includes a local pushbutton and LED for functional testing. The pushbutton, connected to RB0, uses the internal pull-up resistor of the PIC and allows manual triggering of alerts or testing of logic. The LED, connected to RC4 through a 330 Ω resistor, provides immediate visual confirmation of the microcontroller's output state. These features are intended only for debugging and do not interface with the hub.

Subsystem Description

This board serves as the Audio and Alert subsystem for the Irri-Gators project, designed by Rylee Wirt. It provides both audible and visual alerts to indicate system status or errors and connects to the main hub through an 8-pin ribbon cable. Each subsystem in the design has its own regulated 5 V power supply, and only the ground line is shared across the ribbon connection. This prevents backfeeding between supplies while maintaining a common signal reference. The schematic includes clear separation between power, control, and signal-conditioning sections to simplify troubleshooting and future expansion.